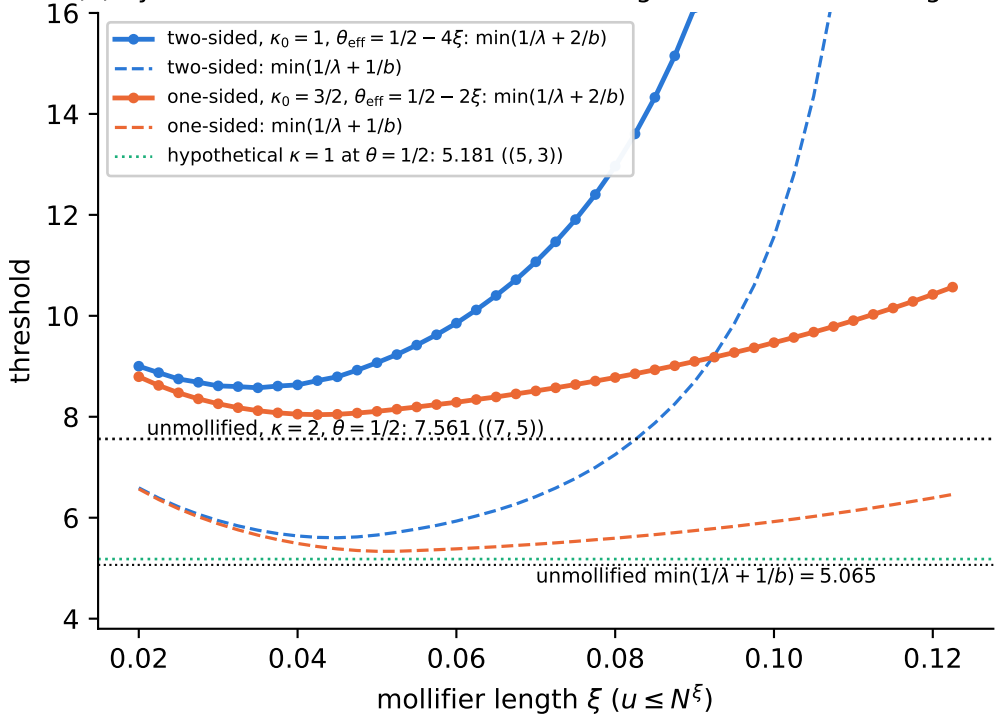
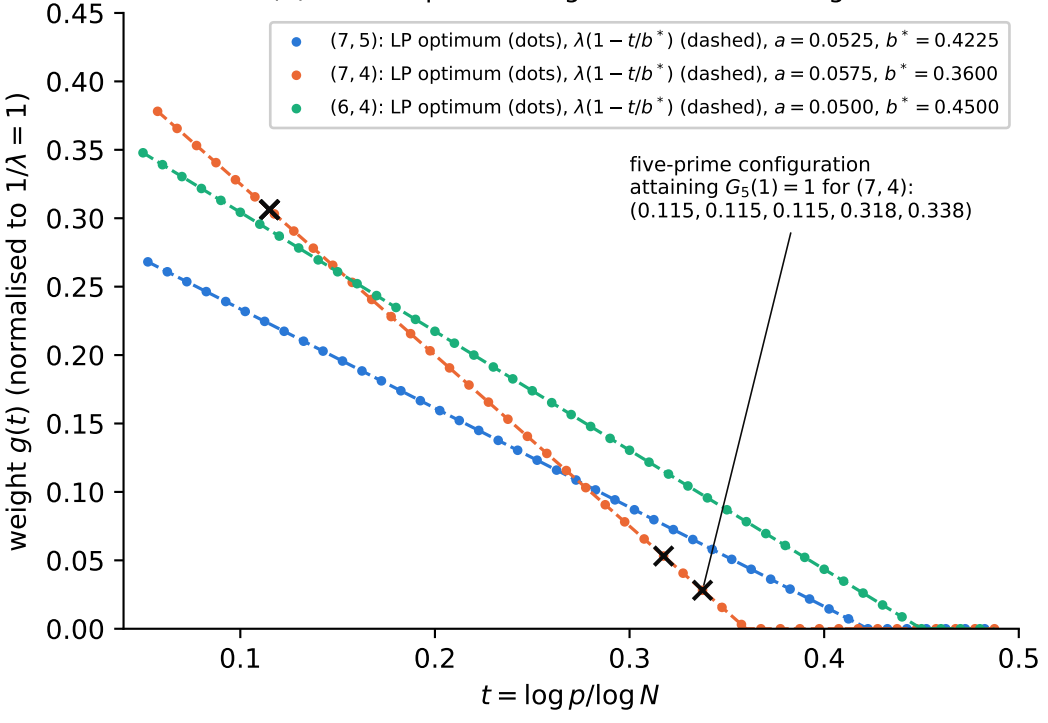

(a) hybrid mollified sieves: the level lost against the dimension gained
two-sided, $\kappa_0 = 1$, $\theta_{\text{eff}} = 1/2 - 4\xi$: $\min(1/\lambda + 2/b)$
two-sided: $\min(1/\lambda + 1/b)$
one-sided, $\kappa_0 = 3/2$, $\theta_{\text{eff}} = 1/2 - 2\xi$: $\min(1/\lambda + 2/b)$
one-sided: $\min(1/\lambda + 1/b)$
hypothetical $\kappa = 1$ at $\theta = 1/2$: 5.181 ((5, 3))
unmollified, $\kappa = 2$, $\theta = 1/2$: 7.561 ((7, 5))
unmollified $\min(1/\lambda + 1/b) = 5.065$
threshold
mollifier length ξ ($u \le N^{\xi}$)
16
14
12
10
8
6
4
0.02
0.04
0.06
0.08
0.10
0.12


(b) the LP-optimal weight is the linear weight
(7, 5): LP optimum (dots), $\lambda(1 - t/b^*)$ (dashed), $a = 0.0525$, $b^* = 0.4225$
(7, 4): LP optimum (dots), $\lambda(1 - t/b^*)$ (dashed), $a = 0.0575$, $b^* = 0.3600$
(6, 4): LP optimum (dots), $\lambda(1 - t/b^*)$ (dashed), $a = 0.0500$, $b^* = 0.4500$
five-prime configuration
attaining $G_5(1) = 1$ for (7, 4):
(0.115, 0.115, 0.115, 0.318, 0.338)
weight $g(t)$ (normalised to $1/\lambda = 1$)
$t = \log p / \log N$
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.1
0.2
0.3
0.4
0.5